Chae-Ghang (Brian) Jeong

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Education

Rice University — Houston, TX
B.S. in Computer Science, Minor in Biosciences — *May 2028 (Expected)*

Experience

Forward Deployed Engineer Intern — May 2026 – Aug. 2026
MakinaRocks — *Seoul, South Korea*

- Engineered conversational AI and backend features for an industrial troubleshooting platform, improving RAG retrieval, implementing deterministic diagnostic scoring, and automating service-report generation.
- Built coordinate-aware parsing for heterogeneous engineering workbooks, substantially improved LLM-based Cause Effect matrix normalization normalization accuracy, and developed automatic header inference and field-level provenance contracts.
- Worked directly with global clients to clarify requirements, present evaluation results and prototypes, propose new AI capabilities, and turn selected ideas into shipped features.

Undergraduate Research — Nov. 2026 – Present
Rice University – ChILi Lab — *Houston, TX*

- Exploring the adaptation of Natural Language Autoencoders (NLAs) to multimodal diffusion models to interpret intermediate activations and investigate how visual concepts and their relationships are represented during image generation.

Projects

Cirqloom | GitHub | *Python, PyTorch, TypeScript, React, Three.js* — 2026

- Built an interactive 3D research platform for exploring GPT-2 Small's 144 attention heads, integrating real attention measurements, controlled experiments, and 26 published IOI circuit heads with evidence-aware visualizations.
- Developed reproducible Python analysis pipelines and versioned cross-language data contracts; implemented attention-pattern detectors, paired controls, and interactive heatmaps to investigate transformer behavior.

StageGround | GitHub | Demo | *Clinical NLP, LLM Evaluation* — 2026

- Designed a clinical LLM evaluation study using 1,000 TCGA pathology reports to investigate whether cancer-stage predictions matching registry labels were supported by source evidence.
- Conducted a 12,000-prediction ablation across four prompting strategies, evaluating label accuracy, abstention, evidence grounding, and semantic support to identify gaps between apparent correctness and source-grounded reliability.

Awards & Certifications

Korea Defense AI Utilization Idea Competition | Top 20 of 300+ Entries — 2025

- Proposed a spatiotemporal infectious-disease risk prediction system using anonymized personnel movement and medical data.

ADsP | Data Analytics Semi-Professional Certification — 2024

Military Service

Republic of Korea Army — Sep. 2024 – Mar. 2026
Sergeant, Mandatory Military Service — *South Korea*

Technical Skills

Languages: Python, TypeScript, C, SQL
AI / ML: PyTorch, Hugging Face, scikit-learn, RAG, LangGraph, LangChain
Backend / Data: FastAPI, Pydantic, Pandas, NumPy, REST APIs
Developer Tools: Docker, Git, pytest